Genetic Parameters and Genome-Wide Association Study of Colostrum Yield and Composition in Holstein Cows

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The genomic architecture of colostrum yield and composition in dairy cattle remains incompletely characterized. This study evaluated phenotypic associations, estimated genetic parameters, and identified genomic regions associated with colostrum yield and composition in Holstein cows. Phenotypic analyses included 4,205 colostrum records from 4,046 cows at the University of Florida Dairy Research Unit and a commercial dairy farm in California; genetic analyses included 3,711 genotyped cows. Mixed models assessed associations with season, parity, gestation length, calf characteristics, prepartum nutrient intake, and subsequent milk production. Genetic parameters were estimated using genomic best linear unbiased prediction. Genome-wide association studies used iterative weighted genomic best linear unbiased prediction, with 1-Mb windows explaining more than 0.3% of additive genetic variance selected for annotation and functional enrichment. Colostrum yield was greater in spring and summer, whereas IgG concentration was greater in fall and winter. Longer gestation and greater calf birth weight were associated with greater colostrum yield. Prepartum dry matter, energy, and metabolizable protein intakes were not associated with colostrum yield or IgG concentration. Colostrum yield was positively associated with milk production during the first 10 weeks of lactation. Heritabilities were 0.30, 0.30, 0.36, 0.29, and 0.26 for colostrum yield, fat concentration, protein concentration, IgG concentration, and total IgG yield, respectively. Genomic regions were identified for all five traits, with shared candidate genes including SLC4A4, BMP5, and IGF1R. Together, these findings demonstrate additive genetic variation in colostrum traits and identify genomic regions for further investigation.